Bernoulli-EKF on SE(3):
A Compact Multi-Object Tracker for Mobile Manipulation

Part I — Algorithmic Overview

The system tracks per-object Bernoulli-Gaussian beliefs over rigid body poses on SE(3), conditioned on RGB-D detections from a moving camera mounted on a manipulator. Per frame k it ingests the SLAM base-pose $T_{wb,k}$, the camera-to-base extrinsic $T_{bc,k}$, the gripper-to-base $T_{bg,k}$, the gripper finger width, and a list of SAM2 detections $\{(\mathrm{pid}_d, \mathrm{label}_d, \mathrm{mask}_d, \mathrm{score}_d)\}_d$ projected through depth.

1. State. Each track o carries $(\mu_{bo}, \Sigma_{bo}, r_o, \ell_o, \tau_o)$: $\mu_{bo} \in \mathrm{SE}(3)$ is the object pose in the *base* frame, $\Sigma_{bo} \in \mathbb{R}^{6\times 6}$ is its covariance in the body-frame tangent at μ_{bo}, $r_o \in [0, 1]$ is the existence probability, ℓ_o a label histogram, and τ_o the SAM2 perception id. *Base-frame storage is the canonical choice*: SLAM uncertainty Σ_{wb} never enters the recursion; the world view is the deterministic composition $T_{wo} = T_{wb}\, \mu_{bo}$.

2. Predict. Two motion models, applied per-track: world-static (default, $u_k = T_{wb,k}^{-1} T_{wb,k-1}$, applied via the adjoint to Σ plus per-phase Q) or rigid-attached (held set, $\delta_{bg,k} = T_{bg,k} T_{bg,k-1}^{-1}$).

3. Detection conditioning. Subpart suppression (`det_dedup.py`) absorbs detections whose 3D voxel sets are mostly contained in another's; centroid back-projection drops masks without valid depth.

4. Association. Translation-only Mahalanobis with a χ^2_3 gate, $G_{\mathrm{out},t} = 21.108$, $R_{cam} = (0.02\, m)^2 I$. Hungarian on the gated cost; SAM2-id continuity bonus $\alpha = 4.4$ breaks ties.

5. Update. For matched pairs on the held track, two safety prefilters guard the Joseph step: held-prefilter (centroid-to-T_{we} within 0.25 m) and innovation-clamp (centroid-to-μ_w within 0.20 m). The Joseph step runs on the *translation block only*; the rotation block and cross-cov are left zeroed (a translation-only observation carries no rotation information). This keeps Σ_{bo} PSD across long runs.

6. Birth and prune. A confirm-3 / score-0.30 / border / near-live (default 5 cm; 25 cm against T_{we} for the held oid) chain controls births. Prune when $r < 0.05$, except hold the held oid above $r = 0.5$.

7. Self-merge. Same-label tracks within 5 cm fuse via information sum, EXCEPT pairs in the relation-protected set (scene graph asserts they are distinct). The held seed is always the keeper.

8. Relations. A VLM/REST backend (`relation_orchestrator.py`) is fired at first call, release transitions, new-oid arrivals, and every 90 frames. Its parent matrix is converted to `RelationEdge` records with the convention **parent = supported, child = supporter** (LLM's i is the supporter; we emit (j, i, on)). A per-edge EMA ($\alpha = 0.3$, threshold 0.5) smooths flicker.

9. Held-set expansion. The set carried by the gripper is the closure of {held seed} under in/on edges; after each self-merge, EMA keys are remapped (drop → keep) so the closure stays valid.

Part II — Detailed Elaboration

II.1 State

`pose_update/gaussian_state.py`.

$$\mathcal{S}_o = (\mu_{bo} \in \mathrm{SE}(3),\ \Sigma_{bo} \in \mathbb{R}^{6\times 6}_{\succeq 0},\ r_o \in [0,1],\ \ell_o,\ \tau_o \in \mathbb{Z},\ \text{age},\ \text{frames\_since\_obs}).$$

Body-frame tangent at μ_{bo}: a perturbation $\xi \in \mathfrak{se}(3)$ acts as $\mu' = \mu\ \mathrm{Exp}(\xi)$ (right-multiply). Ordering $\xi = (\xi_\omega; \xi_v)$ — rotation first then translation. The adjoint

$$\mathrm{Ad}(T) = \begin{bmatrix} R & 0 \\ [t]_\times R & R \end{bmatrix}$$

moves a tangent vector through $T = (R, t)$.

II.2 Predict

World-static. A world-static object satisfies $T_{wb,k}\mu_{bo,k} = T_{wb,k-1}\mu_{bo,k-1}$, hence $\mu_{bo,k} = u_k\, \mu_{bo,k-1}$ with $u_k = T_{wb,k}^{-1} T_{wb,k-1}$. Cov:

$$\Sigma_k = \mathrm{Ad}(u_k)\, \Sigma_{k-1}\, \mathrm{Ad}(u_k)^\top + Q_{\text{static}}.$$

Q_{static} schedules on `frames_since_obs`: $1\mathrm{e}{-6} I_6$ (idle, < 5), $1\mathrm{e}{-5} I_6$ (< 50), $1\mathrm{e}{-8} I_6$ thereafter (`ekf_se3.py`).

Note the cross-coupling in $\mathrm{Ad}(u_k)$: the translation block of Σ accumulates $[t(u_k)]_\times P_{rr} [t(u_k)]_\times^\top$ each frame, so for an unobserved track the body-frame translation uncertainty grows in directions *perpendicular to the robot's motion*. This is correct EKF behaviour, not a bug.

Rigid-attached. For tracks in the held set $\mathcal{H}_k$, $\mu_{bo,k} = \delta_{bg,k}\, \mu_{bo,k-1}$ with $\delta_{bg,k} = T_{bg,k} T_{bg,k-1}^{-1}$. The driver applies a phase-aware Q_{manip}:

- Seed during grasping/releasing: $\mathrm{diag}((0.02)^2{\times}3,\ (0.10)^2{\times}3)$ — physical grip-slack.
- Seed during holding: $\mathrm{diag}(2.5\mathrm{e}{-4}{\times}3,\ 1\mathrm{e}{-3}{\times}3)$ — settled grip.
- Transitive members (apples on tray): always the holding row — they don't deform when the gripper closes around the seed.

II.3 Detection conditioning

Subpart suppression (`det_dedup.py`). Voxelise each mask through depth at 2 cm; for ordered pair (a, b) with $|V_b| \le |V_a|$, if $|V_b \cap V_a|/|V_b| > 0.8$, absorb b into a (label histograms merge). Catches SAM2 mask splits / re-prompts.

Centroid validity. $c_d^{\text{cam}} = \frac{1}{|m_d \cap \text{valid-depth}|} \sum_{p \in m_d} \text{back\_project}(p, \text{depth}, K)$. If the mask has no valid-depth pixels, d is dropped (logged in `centroid_dropped`).

II.4 Association

`pose_update/association.py` $\rightarrow$ `hungarian_associate`. For each (o, d):

$$\begin{aligned} t_{bo}^{\text{meas}} &= T_{bc}\big[c_d^{\text{cam}}; 1\big]_{1:3} \\ R_{bo}^{t} &= R_{bc}\, R_{cam}\, R_{bc}^{\top}, \quad R_{cam} = \text{diag}((0.02\,m)^2 \times 3) \\ \nu_{od} &= t_{bo}^{\text{meas}} - \mu_{bo}[1{:}3, 4], \quad S_{od} = P_{tt,o} + R_{bo}^{t} \\ d_{t,od}^2 &= \nu_{od}^{\top} S_{od}^{-1} \nu_{od}. \end{aligned}$$

Cost matrix $C_{od} = d_{t,od}^2 - \alpha\,[\tau_o = \text{pid}_d] + \text{label\_penalty}$ where label_penalty is 0 for matching labels and 10^{12} otherwise (big-M infeasibility). Hungarian minimises $\sum C_{o\sigma(o)}$; matches with $d_t^2 > G_{\text{out},t} = 21.108$ are dropped.

II.5 Update

Held-track prefilters (`visualize_ekf_tracking.py`, matched-update branch). For each accepted match (o, d) with o the held seed:

- Held-prefilter: $\|c_d^w - (T_{wb} T_{bg})_{1:3,4}\|_2 \le 0.25$ m.
- Innovation clamp: $\|c_d^w - (T_{wb} \mu_{bo})_{1:3,4}\|_2 \le 0.20$ m.

A failure marks the detection consumed (so it doesn't re-birth) and skips the EKF step; the rigid-attach predict carries the track.

Joseph step (translation-only), `gaussian_state.py` $\rightarrow$ `update_observation_centroid`. Let $K_{tt} = P_{tt}\, S^{-1}$ (3×3):

$$\begin{aligned} \delta_t^{\text{base}} &= K_{tt}\, \nu \\ \mu_{bo}[1{:}3, 4] &\leftarrow \mu_{bo}[1{:}3, 4] + \delta_t^{\text{base}} \\ P_{tt}^{\text{post}} &= (I - K_{tt})\, P_{tt}\, (I - K_{tt})^{\top} + K_{tt}\, R_{bo}^{t}\, K_{tt}^{\top} \\ \Sigma_{bo}^{\text{post}} &= \text{block}-\text{diag}\big(P_{tt}^{\text{post}},\ P_{rr}\big). \end{aligned}$$

The cross-covariance is zeroed: a translation-only observation provides no information about rotation (we set $R_{rot} = \infty$), so the honest posterior is block-diagonal. An earlier version preserved the prior cross-cov, which broke $P_{rr} - P_{rt}^{\top} P_{tt}^{-1} P_{rt} \succeq 0$ over many updates; the refactored version is PSD by construction, with an `eigh`-clip belt-and-suspenders.

For pairs where ICP succeeded (`icp_pose.py`), an alternative full-SE(3) update via `joseph_update` in `object_belief.py` replaces the centroid path.

II.6 Birth and prune

Birth gates (`orchestrator.py` $\rightarrow$ `birth_admissible` and `birth_gating.py` $\rightarrow$ `is_near_live_track`):

1. Border: bbox not within `birth_border_margin_px`=10 px of edge.
2. Confirm-k: pending-birth counter $\geq k = 3$.
3. Score floor: rolling-max score ≥ 0.30.
4. Near-live: world-frame centroid > 0.05 m from any same-label live track. For the held oid the radius switches to 0.25 m and T_{we} is checked as a secondary anchor.

Survivors mint a fresh oid with $r_0 = 0.4$ and run ICP once to seed.

Existence updates: matched $r' = \frac{(1-p_s)+p_s\, p_d\, \mathcal{L}\, r}{1-p_s+p_s\, r}$; missed $r' = \frac{(1-p_d\, p_v)\, r}{1-p_d\, p_v\, r}$, p_v from ray-trace (`visibility.py`).

Prune: $r_o < 0.05$ deletes, EXCEPT the held seed which is floored at 0.5 (survives long held-occlusion).

II.7 Self-merge

`orchestrator.py` $\rightarrow$ `_self_merge_pass`. Inputs: held seed s, protected pairs $\mathcal{P} = \{(\min, \max) : (p, c, \cdot) \in \mathcal{E}\}$.

Algorithm 1 Self-merge pass.

for each pair (o_i, o_j) with $\ell_i = \ell_j$ **do**
  if $(\min(o_i, o_j), \max(o_i, o_j)) \in \mathcal{P}$ **then continue**
  end if
  $d \leftarrow \|\mu_w(o_i) - \mu_w(o_j)\|_2$
  if $d > 0.05$ **then continue**
  end if
  Push (d, d_t^2, o_i, o_j) as candidate.
end for
Sort candidates by d ascending.
for each (d, d_t^2, o_i, o_j) **do**
  if o_i or o_j already absorbed **then continue**
  end if
  Choose keep$\leftarrow s$ if $s \in \{o_i, o_j\}$, else higher-r, else lower oid.
  Information-sum-fuse μ, Σ; merge label histograms / sam2_tau / T_oe / r.
  Append {keep, drop, d, d_t^2} to merges.
end for

The protected-pairs gate is what stops an apple-on-tray pair from collapsing once their world centroids drift within 5 cm under occlusion bias.

II.8 Relations

Trigger gate (`relation_utils.py` $\rightarrow$ `should_recompute_relations`): fires at first call, release transitions, new-oid arrivals, or every `relation_every_n_frames`=90 frames. The grasp-transition trigger was removed (LLM at grasp onset has incomplete information).

Backend (`relation_orchestrator.py` + `relation_client.py`). On a triggered frame, the orchestrator builds normalised bboxes per oid (from `det_to_oid`) and calls `LLMRelationClient` or `RESTRelationClient`. The backend returns $p \in [0,1]^{N\times N}$ with p_{ij} = "j rests on i". Disk cache (`CachedRelationClient`) keyed by (frame, N, rounded bboxes).

Convention swap. Held-set expansion uses parent=supported, child=supporter. The LLM's (i,j) encodes the opposite (j rests on i). Edge emission swaps:

$$\text{edge} = \text{RelationEdge}(\text{parent} = j, \text{child} = i, \text{type} = \text{on}, \text{score} = p_{ij}).$$

A regression here was responsible for held-set expansion failing — the closure walks edges with child $\in S$, so a backward emission means no propagation.

EMA (`orchestrator.py` $\rightarrow$ `RelationFilter`). For each key k = (parent, child, type):

$$\text{ema}_k(t) = \alpha\,\text{raw}_k(t) + (1-\alpha)\,\text{ema}_k(t-1), \quad \alpha = 0.3.$$

Edges emitted when ema ≥ 0.5; pruned when ema ≤ 0.01. Half-life $\sim$ 2 trigger calls $\sim$ 180 frames.

Remap after merges. When self-merge produces $\{(\text{drop}_i, \text{keep}_i)\}$, every EMA key and emitted edge has parent/child rewritten via the closure of `drop_to_keep`. Self-loops drop, duplicate keys merge by max.

II.9 Held-set expansion

`relation_utils.py` $\rightarrow$ `expand_held_with_relations`.

$S \leftarrow \{s\}$
for iter $= 1 \ldots 8$ **do**
    changed $\leftarrow$ false
    for each edge $(p, c, t) \in \mathcal{E},\ t \in \{\text{in}, \text{on}\}$ **do**
        if $c \in S$ and $p \notin S$ **then** $S \leftarrow S \cup \{p\}$; changed$\leftarrow$true
        end if
    end for
    if not changed **then break**
    end if
end for
return S

The caller post-filters S against currently-live oids to drop references to pruned tracks the EMA hasn't yet decayed.

II.10 Gripper phase FSM

`pose_update/gripper_state.py` $\rightarrow$ `GripperPhaseTracker`. Four phases: Idle $\rightarrow$ Grasping $\rightarrow$ Holding $\rightarrow$ Releasing $\rightarrow$ Idle. Driven by:

- Width threshold crossings with hysteresis (closed < 0.025 m, open ≥ 0.040 m).
- Stability test (last 5 widths span < 0.01 m).
- Min-frames timer (5) holding the transition phases through brief instability.

At the Idle→Grasping edge, the held oid is selected by `GraspOwnerDetector`: rank detections by inside- gripper-volume containment fraction (= count_inside / total_pts); tiebreak by smallest mask area. Falls back to nearest-live-track within 0.30 m of T_{we}. Self-merges propagate via `apply_merges`: drop = `held_obj_id` → switch to the keep.

II.11 Frame-by-frame algorithm

Algorithm 2 One frame of the Bernoulli-EKF tracker (driver loop).

Inputs: T_{wb}, T_{bc}, T_{bg}, w, RGB, depth, detections.
$g \leftarrow$ GripperPhaseTracker.step$(w, \ldots)$
relation_pipeline.maybe_update$(\ldots)$
$\mathcal{H} \leftarrow$ Expand$(g.\text{held}, \mathcal{E}_t) \cap$ live oids
predict_static(skip $\mathcal{H}$); rigid_attachment_predict(each $o \in \mathcal{H}$)
suppress_subpart_detections; centroid_cam_from_mask; gather `dets_for_assoc`
hungarian_associate → matches, unmatched_tracks, unmatched_dets
for each match (o, d) **do**
    if $o = g$.held and (held-prefilter or innov-clamp triggers) **then** skip update
    end if
    update_observation_centroid; update r_o via match log-likelihood
end for
for each unmatched track o **do**
    Update r_o via miss equation; floor at `r_held_floor` if $o = g$.held
end for
for each unmatched detection d **do**
    if birth_admissible(d) and not is_near_live_track(d) **then**
        Mint new oid, run ICP once, set $r_0 = 0.4$
    end if
end for
Prune oids with $r_o < r_{\min}$ (skip held seed)
$\mathcal{P} \leftarrow \{(\min(p, c), \max(p, c)) : (p, c, \cdot) \in \mathcal{E}_t\}$
_self_merge_pass(held = g.held, protected = $\mathcal{P}$)
g.apply_merges(merges); relation_pipeline.remap_after_merges(merges)

II.12 Numerical defaults summary

R_{cam}	$\mathrm{diag}((0.02\,m)^2)$	centroid measurement
Q_{static}	1e−6 → 1e−5 → 1e−8	schedule on `frames_since_obs`
$Q_{\text{manip seed}}$ (graspg./relsg.)	$\mathrm{diag}((0.02)^2{\times}3, (0.10)^2{\times}3)$	per frame
$Q_{\text{manip seed}}$ (holdg.)	diag(2.5e−4×3, 1e−3×3)	per frame
$Q_{\text{manip transit}}$	holding row above	apples-on-tray
$G_{\text{out},t}$	21.108	χ^2_3 at 99.99%
α_{SAM2}	4.4	continuity bonus
`birth_min_dist_m`	0.05	default near-live radius
`held_birth_radius_m`	0.25	wider for held oid
`held_meas_radius_m`	0.25	T_{we} prefilter
`held_meas_innov_max_m`	0.20	innov clamp
`r_min`	0.05	prune threshold
`r_held_floor`	0.50	held existence floor
`birth_confirm_k`	3	pending-birth count
`birth_score_min`	0.30	rolling-max floor
`self_merge_trans_m`	0.05	Euclidean merge gate
EMA α, threshold	0.3, 0.5	relation smoothing
`relation_every_n_frames`	90	periodic LLM tick

II.13 Implementation index

- State / predict: `pose_update/gaussian_state.py`, `pose_update/object_belief.py`, `pose_update/ekf_se3`
- Detection conditioning: `pose_update/det_dedup.py`, `pose_update/icp_pose.py`
- Association / update: `pose_update/association.py`, `pose_update/orchestrator.py`, `gaussian_state.py`
- Birth / prune: `pose_update/orchestrator.py::birth_admissible`, `pose_update/birth_gating.py`
- Self-merge: `pose_update/orchestrator.py::_self_merge_pass`
- Relations: `pose_update/relation_orchestrator.py`, `pose_update/relation_client.py`, `pose_update/relation_utils.py`
- Gripper FSM: `pose_update/gripper_state.py`, `pose_update/grasp_owner_detector.py`
- Visibility: `pose_update/visibility.py`
- Driver / harness: `tests/visualize_ekf_tracking.py`